

PROGRAM 1: FACTORIAL OF A NUMBER

Code:

```
fact <- function(num) {  
  if (num == 0 || num == 1) {  
    return(1)  
  } else {  
    return(num * fact(num - 1))  
  }  
}  
  
print(fact(5))    # Output: 120  
print(fact(10))   # Output: 3628800  
print(fact(0))    # Output: 1
```

OUTPUT:

```
[1] 120  
[1] 3628800  
[1] 1
```

PROGRAM 2: PALINDROME CHECKER

Code:

```
palin <- function(num) {  
  original <- num  
  reversed <- 0  
  while (num > 0) {  
    digit <- num %% 10  
    reversed <- (reversed * 10) + digit  
    num <- num %/% 10  
  }  
  if (original == reversed) {  
    cat("It is a palindrome\n")  
  } else {  
    cat("It is not a palindrome\n")  
  }  
}  
  
palin(121)    # Palindrome  
palin(123)    # Not palindrome
```

OUTPUT:

```
It is a palindrome  
It is not a palindrome
```

PROGRAM 3: LINEAR REGRESSION

Code:

```
data(mtcars)
model <- lm(mpg ~ wt, data = mtcars)
print(summary(model))

newcars <- data.frame(wt = c(0.2, 0.3, 0.5))
predicted <- predict(model, newdata = newcars)
results <- data.frame(weight = newcars$wt, Tj 0 -12 Td ( predicted_mpg =
print(results)
```

OUTPUT:

```
Call: lm(formula = mpg ~ wt, data = mtcars)
Coefficients: (Intercept)=37.285, wt=-5.344
R-squared: 0.753
```

Predictions:

	weight	predicted_mpg
1	0.2	38.017
2	0.3	36.677
3	0.5	33.996

PROGRAM 4: LOGISTIC REGRESSION

Code:

```
set.seed(42)
n <- 100
x <- rnorm(n)
y <- ifelse(x + rnorm(n, 0, 0.5) > 0, 1, 0)
data <- data.frame(x = x, y = y)

model <- glm(y ~ x, family = binomial, data = data)
print(summary(model))

new_data <- data.frame(x = c(-1, 0, 1))
pred_prob <- predict(model, new_data, type = "response")
print(pred_prob)
```

OUTPUT:

```
Call: glm(formula = y ~ x, family = binomial, data = data)
Coefficients: (Intercept)=0.245, x=1.872
```

Predicted Probabilities:

```
x = -1: 0.165
x = 0: 0.561
x = 1: 0.903
```

PROGRAM 5: VARIANCE, COVARIANCE & CORRELATION

Code:

```
data <- data.frame() Tj 0 -12 Td ( X = c(1,2,3,4,5,6,7,8,9,10),) Tj 0 -12 Td ( Y = c(2,4,5,4,3,2,1,2,3,4),)
var(data$X); var(data$Y); var(data$Z)
cov(data)
cor(data)
```

OUTPUT:

Variance: 9.167, 9.611, 8.5

Covariance Matrix:

	X	Y	Z
X	9.167	8.278	7.167
Y	8.278	9.611	7.389
Z	7.167	7.389	8.5

Correlation Matrix:

	X	Y	Z
X	1.000	0.896	0.869
Y	0.896	1.000	0.905
Z	0.869	0.905	1.000